STEP 3 — Data Import Karna

Time: ~10 min · Is step ka output: `data/raw/event_log.csv` taiyaar

> **Aapke project ke liye 2 raaste hain:**
> **Raasta A (easy):** project khud ka synthetic hospital data banata hai — `run_all.py` chalaiye, kaam ho gaya.
> **Raasta B (asli data):** apna CSV daaliye aur `python run_all.py --skip-data` chalaiye.

3.1 Event log kya hota hai — 30 second me samjho

Process mining ka fuel **event log** hai. Ek hospital event log ka har row ek **ghatna (event)** batata hai: “kis patient ke saath, kya kaam hua, aur kab hua”.

```
case_id     activity                 timestamp            resource      department
CASE-00001  Registration             2026-03-01 09:14:00  REC-01        Front Desk
CASE-00001  Triage                   2026-03-01 09:31:00  NUR-EM-01     Emergency
CASE-00001  Specialist Consultation  2026-03-01 10:12:00  DR-OP-02      Outpatient
CASE-00001  Lab Test                 2026-03-01 11:40:00  LAB-TECH-01   Laboratory
CASE-00001  Diagnosis Confirmed      2026-03-01 12:25:00  DR-OP-02      Outpatient
CASE-00002  Registration             2026-03-01 10:02:00  REC-03        Front Desk
```

Ek **case** = ek patient ka poora journey. Upar wala `CASE-00001` ke saare rows milkar ek complete pathway banate hain.

3.2 Column specification (yeh 3 columns MUST hain)

Column	Mandatory?	Matlab	Example
`case_id`	**YES**	Patient journey ki unique id	`CASE-00001`
`activity`	**YES**	Jo step hua	`Lab Test`
`timestamp`	**YES**	Kab hua (date + time dono)	`2026-03-01 09:14:00`
`resource`	Optional	Doctor / nurse / machine	`LAB-TECH-01`
`department`	Optional	Department ka naam	`Laboratory`
`wait_minutes`	Optional	Step se pehle kitna wait hua	`42.5`
`service_minutes`	Optional	Step me kitna time laga	`51.2`
`cost_inr`	Optional	Us step ka kharcha	`1536.00`

> Optional columns nahi honge to bhi project chalega — lekin cost chart aur resource workload chart khali aayenge. Ho sake to **kam se kam** `resource` **zaroor daaliye**, warna report me “resource analysis” section kamzor lagega.

3.3 Raasta A — synthetic data (default, sabse aasan)

Kuch bhi karne ki zarurat nahi. Jab aap `python run_all.py` chalayenge, Step 1 apne aap 1,500 patients ka realistic event log bana dega.

```
python src/generate_data.py
```

Output:

```
13:11:33 | INFO | careflow.generate_data | events=13664  cases=1500  activities=15
13:11:33 | INFO | careflow.generate_data | saved data/raw/event_log.csv
```

Banegi **15 clinical activities** wali 4 pathways:

Pathway	Weight	Activities (sequence)
P1 Outpatient Standard	35 %	Registration → Triage → Specialist Consultation → Lab Test → Diagnosis Confirmed → Medication Prescribed → Discharge
P2 Emergency Surgical	25 %	Registration → Triage → Emergency Assessment → Lab Test → Radiology → Diagnosis → Admission → Surgery → Observation → Physiotherapy → Discharge Planning → Discharge
P3 Chronic Followup	25 %	Registration → Specialist Consultation → Lab Test → Medication → Physiotherapy → Discharge Planning → Discharge
P4 ICU Escalation	15 %	Registration → Triage → Emergency Assessment → Lab Test → ICU Transfer → Observation → Radiology → Diagnosis → Medication → Discharge Planning → Discharge

Data me jaan-boojh kar 2 asli duniya ki galtiyan daali gayi hain, taaki aap cleaning dikha sakein:

- **0.6 % duplicate events** (same patient ka same step do baar record ho gaya)
- **1.5 % rows me resource missing** (kis staff member ne kiya, woh blank)
- **Rework loops** — Lab Test, Radiology, Physiotherapy dobara ho jaate hain (hospital me aam hai)

Data ko apni marzi se badalna

`config/config.yaml` kholiye aur `data_generation` section edit kijiye:

```
data_generation:
  n_cases: 1500                # kitne patients (3000 kar dijiye to bada data)
  seed: 42                     # isse number badalne par naya data banega
  start_date: "2026-03-01"     # data ka pehla din
  end_date: "2026-08-31"       # data ka aakhri din
  duplicate_event_rate: 0.006  # duplicate galti ka rate
  missing_resource_rate: 0.015 # blank resource ka rate
```

```
python src/generate_data.py --cases 3000 --seed 7
```

3.4 Raasta B — apna asli data import karna

Qadam 1 — CSV file taiyaar kijiye

Excel / Google Sheets me apna data kholiye. Column names bilkul aise rakhiye (chhote letters, underscore, space nahi):

```
case_id, activity, timestamp, resource, department, wait_minutes, service_minutes, cost_inr
```

Column ka order matter nahi karta. Extra columns bhi rakh sakte hain — project unhe ignore kar dega.

Qadam 2 — timestamp format theek kijiye

Yeh format use kijiye: **`YYYY-MM-DD HH:MM:SS`** jaise `2026-03-01 09:14:00`

- Excel me date `01/03/2026` likhi ho to usse format kijiye
- Sirf date ho to time `00:00:00` jod dijiye
- Excel me date ek number (jaise `46082`) dikhe to cell format "Text" karke `2026-03-01 09:14:00` type kijiye

Qadam 3 — file sahi jagah rakhiye

```
CareFlow/data/raw/event_log.csv
```

File ka naam **bilkul** `event_log.csv` hona chahiye. Windows kabhi-kabhi `event_log.csv.csv` bana deta hai (kyunki "file extensions hide" hoti hain) — Explorer me View → "File name extensions" on karke check kar lijiye.

Qadam 4 — optional: patient ki details wali file

Agar aapke paas patient-level attributes hain to unhe yahan rakhiye: `data/raw/case_attributes.csv`

```
case_id, patient_id, age, gender, triage_level, admission_type, pathway, los_hours,
total_cost_inr
CASE-00001, PAT-1001, 67, Male, 2, Emergency, P2_Emergency_Surgical, 31.5, 48210.00
```

Yeh file **optional** hai. Na dein to project khud calculate kar lega.

Qadam 5 — pipeline chalao (BINA naya data banaye)

```
python run_all.py --skip-data
```

`--skip-data` ka matlab: "naya data mat banao, jo mere paas hai wahi use karo".

3.5 Real duniya ke datasets (agar aap public data use karna chahein)

Dataset	Kya hai	Kahan se

BPI Challenge 2015	Asli Dutch municipality log — process mining ki sabse jaani-maani dataset	`research.tue.nl` (4TU.ResearchData) aur `processmining.org/event-data.html`
BPI Challenge 2011–2020	Alag-alag industry ke asli event logs	`icpmconference.org` (BPI Challenge pages)

> Yeh public logs **healthcare ke nahi** hain (municipality / finance / manufacturing ke hain). CareFlow me daal sakte hain, lekin `activity` ke naam hospital wale nahi rahenge. Agar mentor hospital data maangta hai to synthetic data hi theek hai — aur report me “synthetic data generated per assignment confidentiality constraints” likh dijiye.

3.6 Import verification — ye 5 check kijiye

```
python -c "
import pandas as pd
df = pd.read_csv('data/raw/event_log.csv')
print('rows          :', len(df))
print('cases         :', df['case_id'].nunique())
print('activities    :', df['activity'].nunique())
print('columns       :', list(df.columns))
print('date range    :', df['timestamp'].min(), '->', df['timestamp'].max())
print('missing rows  :', df.isna().sum().sum())
print(df.head(3))
"
```

Expected output (synthetic data ke saath):

```
rows          : 13664
cases         : 1500
activities    : 15
columns       : ['case_id', 'activity', 'timestamp', 'resource', 'department',
                 'wait_minutes', 'service_minutes', 'cost_inr']
date range    : 2026-03-01 07:26:00 -> 2026-08-31 05:46:50
missing rows  : 203
```

Verification checklist

Check	Expected	Fail hone par
`case_id` column exists	Haan	Column rename kijiye
`activity` column exists	Haan	Column rename kijiye
`timestamp` column exists	Haan	Column rename kijiye
Cases ki count	> 50	case_id galat column par hai
Activities ki count	> 5	activity galat column par hai
Timestamp range	6 months	timestamp format theek kijiye

Data ready hai! Ab Step 4 — poora pipeline chalana.